

Prescribed Finite Étale Algebras as Full Fibers of Keller Maps with Symmetric Monodromy

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Abstract

Every prescribed finite étale K -scheme of rank $N \geq 3$ is the complete full fiber of an explicit polynomial Keller map $F : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$, preserving its entire arithmetic structure. Here K has characteristic zero, $\det DF = 1$, and $\text{gdeg}(F) = N$. Given a squarefree presentation $A \simeq K[T]/(P)$, the map uses one translation $P(a + S)$ and a uniform quadratic-gauge formula of coordinate degree at most $6N + 2$. The fiber identification represents $\text{Hom}_{K\text{-alg}}(A, -)$, naturally in every commutative test algebra and compatibly with extension of the ground field.

For every seed produced by the construction, not merely for a generic choice of coefficients, the generic inverse equation has geometric and arithmetic Galois group S_N . Its primitive action implies that every realization map is absolutely and stably compositionally indecomposable. Over a Hilbertian field, each fixed map also has infinitely many connected full fibers with splitting-field group S_N . Thus the result prescribes the complete fiber algebra, rather than only producing maps of a chosen generic degree.

1 Introduction and main results

A polynomial map with nonzero constant Jacobian is étale. Its fibers are therefore finite étale schemes, possibly empty. This raises a simple reverse question: given a finite étale algebra A/K , can we build a Keller map with $\text{Spec } A$ as one of its fibers?

There is one important word in that question. The fiber should be *full*: its length must equal the geometric degree of the map. Otherwise some generic inverse sheets may have escaped to infinity, and the special fiber records only part of the inverse problem. Our main result answers the prescribed-algebra question. Every finite étale K -algebra of rank at least three occurs as a full Keller fiber in three variables. The construction is explicit once a squarefree presentation $A \simeq K[T]/(P)$ and an admissible translation parameter are supplied. For an abstract algebra, monogenicity supplies such a presentation existentially, but not canonically.

For precision, let K be a field of characteristic zero. A polynomial map

$$F : \mathbb{A}_K^m \longrightarrow \mathbb{A}_K^m$$

is a *Keller map* if $\det DF \in K^*$, and its geometric degree is

$$\text{gdeg}(F) = [K(x_1, \dots, x_m) : K(F_1, \dots, F_m)].$$

Such a map is étale and hence quasi-finite, so every fiber over a K -point is a finite étale K -scheme, possibly empty.

Definition 1.1 (Keller fibers and fullness). *If $F : \mathbb{A}_K^m \rightarrow \mathbb{A}_K^m$ is a Keller map and $y \in \mathbb{A}^m(K)$, then $F^{-1}(y)$ is a Keller fiber. It is a full Keller fiber if*

$$\text{rank}_K F^{-1}(y) = \text{gdeg}(F).$$

Here $\text{rank}_K X = \dim_K \Gamma(X, \mathcal{O}_X)$ for a finite K -scheme X . Thus “Keller fiber” means any ordinary fiber; the adjective “full” says that the special fiber contains every generic inverse sheet. A shorter fiber can occur when sheets escape through the nonproperness boundary.

Producing Keller maps in each generic degree is now shared terrain. Gallagher gives explicit all-degree families, and Migus explicitly takes the available all-degree constructions as input in his classification of generic degrees for real Keller maps with non-dense image [3, 9]. Shaska develops the mixed-sign grading and quotient geometry of the announced cubic map [11]. These results clarify why large families should exist. They do not address the reversed quantifier order here: begin with an arbitrary finite étale algebra, then construct a map whose *complete* fiber represents that algebra.

For rank $N \geq 3$, choose a squarefree presentation

$$A \simeq K[T]/(P)$$

and $a \in K$ with $P'(a)P'''(a) \neq 0$, and put $G(S) = P(a + S) - P(a)$. The proof then has five steps.

1. Design the scalar inverse equation

$$E_{\Pi, B, C}(S) = G_{\Pi}(S) - \frac{g_1}{2}(BS^2 + C).$$

2. Arrange the exact derivative identity

$$E'_{\Pi, B, C}(S) = g_1 D, \quad D = 1 - SQ + \Pi S^2.$$

3. Use the unit D to reconstruct the source rationally from a simple root S ; the complete formula is boxed in Section 2.
4. Choose the quadratic gauge so that this reconstruction polynomializes in (x, y, z) , while the source and target Jacobian factors D^{-1} and $-2D$ cancel.
5. At the selected target, specialize E to $P(a + S)$. Squarefreeness makes E' invertible modulo E , so the localization disappears and the full fiber is $\text{Spec } K[T]/(P)$.

A fixed linear change in the target makes the Jacobian one, and all coordinate degrees are at most $6N + 2$. The detailed reconstruction is natural over arbitrary commutative test algebras; it is this derivative-unit argument, rather than the graded quotient alone, that identifies the exact scheme fiber.

Together with the elementary fact that every Keller fiber is finite étale, the realization theorem gives an exact fixed-degree classification. For $N \geq 1$, a finite K -scheme of rank N is a full Keller fiber of a geometric-degree- N map in $\mathbb{A}_K^3$ exactly when it is finite étale and $N \neq 2$.

The same coordinates identify the actual function-field extension. On the dense chart $1 + xy \neq 0$, the source field is

$$K(x, y, z) = K(\Pi, B, S),$$

and S has minimal polynomial $E_{\Pi, B, C}$ over $K(\Pi, B, C)$. Since that polynomial is irreducible of degree N , the displayed map has geometric degree N . This comparison is what turns the literal rank- N fiber into a full Keller fiber.

The generic inverse problem is also as large as possible in its Galois theory. For every admissible degree- N seed, not just a generic choice of its coefficients, the geometric and arithmetic Galois groups of $E_{\Pi,B,C}$ are both S_N . The proof fixes Π , parametrizes the repeated-root curve, and shows that all but finitely many vertical slices are Morse. The classical Morse-polynomial theorem gives S_N on one slice, and the finite-étale ordered-root cover embeds that group into the generic geometric group. Over a Hilbertian field, the same fixed map consequently has infinitely many connected full Keller fibers with splitting-field group S_N . Since the natural action of S_N is primitive, there is no proper intermediate generic function field. The chain rule then turns this into a structural consequence: every realization map remains compositionally indecomposable after every characteristic-zero extension and every stabilization by identity variables.

The same inverse equation also detects precisely where affine sheets are lost. The logically separate calculation is placed in Appendix D. Over an algebraic closure, the reduced nonproperness locus is

$$V_{\text{red}}(\text{Disc}_S E_{\Pi,B,C}).$$

This statement is set-theoretic, or equivalently an equality of reduced closed subschemes; it does not identify the generally nonreduced discriminant scheme with a scheme structure on the nonproperness locus. The slice $\Pi = 0$ is treated separately by direct source charts and a Newton polygon calculation.

The reconstruction is stronger than a count of geometric points. Its formulas are natural in the test algebra, so they identify the whole fiber functor, including nilpotents and every scalar extension. After translating back from S to T , the literal fiber is the original algebra $K[T]/(P)$. Nothing arithmetic has been replaced by a numerical proxy.

The statements are separated according to their dependencies. The polynomial realization theorem starts from the supplied pair (P, a) and records the map, its determinant, geometric degree, and exact fiber; the immediately following effective corollary records the degree bound and naturality. Monogenicity then gives the finite-étale realization corollary. Symmetric monodromy is a separate theorem about the generic inverse equation, followed by the stable-atomicity corollary. The rank classification combines realization with the classical degree-two obstruction; its scalar-extension and descent argument is recorded in Appendix B. The nonproperness calculation, which is logically independent of this hierarchy, is collected in Appendix D; the coefficientwise pullback calculation is kept separately in Appendix A.

Arithmetic consequences remain literal transfers: field decomposition, Galois action, signatures, local points, and rational points are preserved because the whole finite étale algebra is preserved. Concrete arithmetic applications are kept in separate certificates rather than developed in this paper.

The polynomial-presentation construction and its passage to abstract finite étale algebras in characteristic zero have been formalized in Lean. Starting from a squarefree polynomial of degree at least three, the formalization chooses an admissible translation, constructs the Jacobian-one map and target, proves geometric degree N and the $6N + 2$ coordinate bound, and identifies the full Keller fiber naturally with $\text{Spec } K[T]/(P)$, including its rank and finite étaleness. It also noncomputably constructs a squarefree polynomial presentation of every abstract finite étale K -algebra and transports the full natural fiber certificate across that presentation. Appendix C records the correspondence between these paper statements and their Lean declarations.

2 The quadratic gauge from the inverse equation

The polynomial realization theorem rests on one explicit construction. Let

$$G(S) = g_1 S + g_2 S^2 + \cdots + g_N S^N \in K[S], \quad g_1 g_3 g_N \neq 0. \quad (2.1)$$

The construction is easiest to read backwards, from the inverse problem. Begin with birational coordinates (Π, S, Q) and put

$$D = 1 - SQ + \Pi S^2. \quad (2.2)$$

For the prescribed one-variable seed, define

$$G_\Pi(S) = g_1 S + \Pi(g_2 S^2 + g_3 S^3) + \sum_{k=4}^N g_k \Pi^k S^k \quad (2.3)$$

and

$$\beta(\Pi, S) = \frac{G'_\Pi(S)/g_1 - 1 - \Pi S^2}{S}. \quad (2.4)$$

The numerator is divisible by S , so β is a polynomial. We now choose the second target coordinate to be

$$B = Q + \beta(\Pi, S). \quad (2.5)$$

This choice is the whole gauge. Indeed,

$$D = 1 - S(B - \beta(\Pi, S)) + \Pi S^2 = \frac{G'_\Pi(S)}{g_1} - BS.$$

Define the third target coordinate by

$$C = \frac{2G_\Pi(S)}{g_1} - BS^2. \quad (2.6)$$

Then the inverse equation

$$E_{\Pi,B,C}(S) = G_\Pi(S) - \frac{g_1}{2}(BS^2 + C) = 0 \quad (2.7)$$

holds by construction, and its derivative satisfies

$$D = \frac{G'_\Pi(S)}{g_1} - BS = \frac{E'_{\Pi,B,C}(S)}{g_1}. \quad (2.8)$$

This identity already gives the root-to-source reconstruction. Fix $(\pi, b, c) \in K^* \times K^2$, let s be a root of $E_{\pi,b,c}$, and suppose $E'_{\pi,b,c}(s)$ is invertible. Then the source point is recovered by the single formula

$$\boxed{\begin{aligned} d &= \frac{E'_{\pi,b,c}(s)}{g_1}, & Q &= b - \beta(\pi, s), & t &= d^{-1}, \\ x &= sd^{-1}, & y &= Q - \pi s, & q &= \pi d, \\ z &= d^2 \left(q - \frac{g_1}{g_3} y^2 (1 + 3t) \right). \end{aligned}} \quad (2.9)$$

The only localization is the inverse of the derivative unit d . The proof of Theorem 3.2 verifies over an arbitrary commutative test algebra that this formula and extraction of the marked root are mutually inverse.

The Jacobian calculation is already visible in these coordinates. At fixed Π , differentiating (2.5)–(2.6) gives

$$\det \frac{\partial(B, C)}{\partial(S, Q)} = -2D. \quad (2.10)$$

Thus any birational source chart with Jacobian D^{-1} will produce a constant Jacobian -2 . What remains is to find such a chart for which B and C pull back to polynomials.

2.1 The polynomial pullback

In source coordinates (x, y, z) , set

$$t = 1 + xy, \quad q = t^2 z + \frac{g_1}{g_3} y^2 (1 + 3t). \quad (2.11)$$

On $t \neq 0$, define

$$\Pi = tq, \quad S = \frac{x}{t}, \quad Q = y + xq. \quad (2.12)$$

Then

$$D = 1 - SQ + \Pi S^2 = t^{-1},$$

and direct differentiation gives

$$\det \frac{\partial(\Pi, S, Q)}{\partial(x, y, z)} = D^{-1}. \quad (2.13)$$

Consequently, the two Jacobian factors are

$$D^{-1} \cdot (-2D) = -2.$$

The particular polynomial q in (2.11) is chosen so that the compact coordinates (2.5)–(2.6) have no denominators after this substitution. Expanding their pullbacks gives the polynomial map $F_G = (\Pi, B, C) : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$:

$$\begin{aligned} \Pi &= tq, \\ B &= y + 3\frac{g_3}{g_1}xq + 2\frac{g_2}{g_1}tq + \sum_{k=4}^N k \frac{g_k}{g_1} t^2 x^{k-2} q^k, \\ C &= x(5 - 3t) - \frac{g_3}{g_1} x^3 z - \sum_{k=4}^N (k-2) \frac{g_k}{g_1} (xq)^k. \end{aligned} \quad (2.14)$$

The long formula is therefore the polynomial pullback of the simple inverse problem above, not a formula guessed in the source coordinates. Once (2.11) is fixed, the pullback is forced coefficient by coefficient. Appendix A records that expansion in full.

2.2 Motivation: the cubic seed and its quotient

For $G(S) = S + S^3$, formula (2.14) specializes coordinatewise to the map announced by Alpöge [1]. We use that map only as the cubic seed of the construction; its determinant and collision now also have an independent Isabelle/HOL formalization [4]. The geometric motivation is normalized multiplication of a marked linear factor and a quadratic factor of a binary cubic: the three marked roots give the three inverse branches, while a nonzero resultant supplies local invertibility [13]. Shaska's mixed-sign grading and quotient viewpoint gives a complementary explanation of the Jacobian cancellation on the contracted quotient [11]. Neither viewpoint replaces the derivative-unit formula (2.9), which is what identifies the complete fiber functor for an arbitrary inverse polynomial.

Thus the three-variable inverse problem is governed by the single equation $E_{\Pi, B, C}(S) = 0$, and the same derivative that controls reconstruction is exactly the factor cancelled in the Jacobian.

3 Localized fiber reconstruction and geometric degree

For $(\pi, b, c) \in K^* \times K^2$, write $\mathcal{F}_{\pi,b,c}$ for the fiber functor

$$\mathcal{F}_{\pi,b,c}(A) = \{(x, y, z) \in A^3 : F_G(x, y, z) = (\pi, b, c)\}$$

on commutative K -algebras.

Lemma 3.1 (Derivative units over test algebras). *Let $E \in K[S]$ be separable, let A be a commutative K -algebra, and let $s \in A$ satisfy $E(s) = 0$. Then $E'(s)$ is a unit of A .*

Proof. Choose $U, V \in K[S]$ with $UE + VE' = 1$. Evaluating at s gives $V(s)E'(s) = 1$. Since A is commutative, this is a two-sided inverse. $\square$

Theorem 3.2 (General localized quadratic-gauge fiber theorem). *The map F_G satisfies*

$$\det DF_G = -2, \quad \text{gdeg}(F_G) = N.$$

Its generic inverse equation is $E_{\Pi,B,C}(S) = 0$. Fix $(\pi, b, c) \in K^ \times K^2$, put $E = E_{\pi,b,c}$, and let $\overline{E'}$ be the class of E' in $K[S]/(E)$. Then*

$$\boxed{F_G^{-1}(\pi, b, c) \simeq \text{Spec}((K[S]/(E))[1/\overline{E'}])}. \quad (3.1)$$

Equivalently, for every commutative K -algebra A there is a natural bijection

$$\text{Hom}_{K\text{-alg}}((K[S]/(E))[1/\overline{E'}], A) \simeq \mathcal{F}_{\pi,b,c}(A). \quad (3.2)$$

Proof. The construction above gives the identities

$$B = Q + \beta(\Pi, S), \quad C = \frac{2G_{\Pi}(S)}{g_1} - BS^2. \quad (3.3)$$

Equations (2.13) and (2.10) give $\det DF_G = -2$ on the dense open set $t \neq 0$, and hence everywhere. For the cubic seed, this agrees with the normalized marked-factor interpretation above. The displayed calculation proves that the same cancellation persists for every admissible seed G .

We now identify the entire fiber. Let A be an arbitrary commutative K -algebra. An A -point (x, y, z) above (π, b, c) satisfies

$$tq = \pi.$$

Because π is a unit, both t and q are units. The chart coordinates

$$S = xt^{-1}, \quad Q = y + xq, \quad D = t^{-1}$$

are therefore defined on every point of the fiber. Equations (3.3) and (2.8) give

$$Q + \beta(\pi, S) = b, \quad E(S) = 0, \quad D = \frac{E'(S)}{g_1}.$$

Thus every source point gives a root S of E for which $E'(S)$ is a unit.

Conversely, suppose $s \in A$ satisfies $E(s) = 0$ and $E'(s) \in A^*$. Put the variables d, Q, t, x, y, q, z equal to the expressions in the boxed reconstruction formula (2.9). The derivative identity says exactly that

$$d = 1 - sQ + \pi s^2.$$

A calculation in the ring A now gives

$$1 + xy = t, \quad t^2 z + \frac{g_1}{g_3} y^2 (1 + 3t) = q, \quad tq = \pi.$$

The first identity in (3.3) gives $B = b$, and the root equation gives $C = c$. We have reconstructed a point of the original fiber.

The two procedures undo one another. Starting with a source point, one recovers

$$D = t^{-1}, \quad x = S/D, \quad y = Q - \pi S, \quad q = \pi D,$$

and then z from the definition of q . Starting with a root, the marked coordinate is again s . All formulas use ring operations and the inverse of the explicitly specified unit $E'(s)$, so they commute with homomorphisms of test algebras.

The universal property of $K[S]/(E)$ identifies its maps to A with roots of E in A . Localizing at $\overline{E'}$ imposes exactly the additional condition that $E'(s)$ be a unit. This proves (3.2), and Yoneda's lemma gives (3.1). Notice that this general localized statement requires no separability hypothesis: it removes precisely the ramified part of the inverse equation.

It remains to check the generic degree. Put

$$L = K(\Pi, B), \quad \lambda = \frac{g_1}{2},$$

and write

$$E_{\Pi, B, C}(S) = H_{\Pi, B}(S) - \lambda C \in L[C][S].$$

This polynomial has degree N in S . Its leading coefficient is $g_3 \Pi$ when $N = 3$ and $g_N \Pi^N$ when $N \geq 4$, so it is a unit of $L[C]$. Hence $E_{\Pi, B, C}$ is primitive over the UFD $L[C]$.

Assume it factors nontrivially over $L(C)[S]$. Gauss's lemma then gives a factorization in $L[C][S]$. Since the polynomial is linear in C , one factor, say $A(S)$, is independent of C . Comparing the coefficient of C shows that $A(S)$ divides the nonzero scalar $-\lambda$ in $L[S]$. Thus A is a unit, a contradiction. Therefore $E_{\Pi, B, C}$ is irreducible of degree N over $K(\Pi, B, C)$.

We finish by comparing that root extension with the actual source field. On the dense chart $t \neq 0$, the change of variables is birational:

$$K(x, y, z) = K(\Pi, S, Q).$$

The forward inclusion follows from the definitions. For the reverse inclusion put $D = 1 - SQ + \Pi S^2$. Then

$$t = D^{-1}, \quad x = \frac{S}{D}, \quad y = Q - \Pi S, \quad q = \Pi D, \quad (3.4)$$

and the defining equation for q recovers z rationally. Since

$$B = Q + \beta(\Pi, S),$$

we have $K(\Pi, S, Q) = K(\Pi, B, S)$. The second marked-line identity gives

$$C = \frac{2G_{\Pi}(S)}{g_1} - BS^2.$$

Because $K(\Pi, B, S) = K(x, y, z)$ has transcendence degree three and is generated by the three elements Π, B, S , those elements are algebraically independent. In particular, over $L_0 = K(\Pi, B)$, the element S is transcendental and the displayed expression for C is a nonconstant degree- N

polynomial in S . Hence C is transcendental over L_0 : otherwise S , which is algebraic over $L_0(C)$, would be algebraic over L_0 . Thus Π, B, C are algebraically independent and $T = K(\Pi, B, C)$ is the actual target function field.

The evaluation homomorphism from the abstract target rational function field to the source field is therefore injective. Under it, $E_{\Pi, B, C}$ has the root S , and the reconstruction above gives

$$K(x, y, z) = K(\Pi, B, S) = T(S).$$

The irreducibility just proved makes $E_{\Pi, B, C}$ the minimal polynomial of S over T . Therefore

$$[K(x, y, z) : K(\Pi, B, C)] = [T(S) : T] = N,$$

which is exactly $\text{gdeg}(F_G) = N$. $\square$

Corollary 3.3 (Finite-étale (squarefree) quadratic-gauge fibers). *In the setting of Theorem 3.2, suppose that $E = E_{\pi, b, c}$ is squarefree of degree N . Then the localization disappears and, for every commutative K -algebra A , there is a natural bijection*

$$\text{Hom}_{K\text{-alg}}(K[S]/(E), A) \simeq \mathcal{F}_{\pi, b, c}(A). \quad (3.5)$$

Consequently,

$$F_G^{-1}(\pi, b, c) \simeq \text{Spec } K[S]/(E),$$

and this is a full Keller fiber of rank N (in particular, finite étale).

Proof. Since E is squarefree, choose $U, V \in K[S]$ with $UE + VE' = 1$. Modulo E , the class of V is an inverse for $\overline{E'}$, so the localization in Theorem 3.2 is just $K[S]/(E)$. The natural equivalence and scheme isomorphism follow from that theorem. Its geometric-degree statement makes the rank- N fiber full. $\square$

Corollary 3.4 (Jacobian-one effective form). *Let*

$$L(\Pi, B, C) = \left(\Pi, -\frac{B}{2}, C \right), \quad \tilde{F}_G = L \circ F_G.$$

Then

$$\det D\tilde{F}_G = 1, \quad \text{gdeg}(\tilde{F}_G) = N.$$

For every $(\pi, b, c) \in K^* \times K^2$,

$$\tilde{F}_G^{-1} \left(\pi, -\frac{b}{2}, c \right) \simeq \text{Spec} \left((K[S]/(E_{\pi, b, c})) [1/\overline{E'_{\pi, b, c}}] \right).$$

If $E_{\pi, b, c}$ is squarefree of degree N , this reduces to

$$\tilde{F}_G^{-1} \left(\pi, -\frac{b}{2}, c \right) \simeq \text{Spec } K[S]/(E_{\pi, b, c}).$$

In particular, a target with $b = 0$ is unchanged by the normalization. Moreover,

$$\deg \Pi \leq 7, \quad \deg B \leq 6N + 2, \quad \deg C \leq 6N,$$

so

$$\max_i \deg(\tilde{F}_G)_i \leq 6N + 2.$$

Proof. The linear map L has determinant $-1/2$, so the determinant, geometric degree, and localized-fiber statements follow from Theorem 3.2; the squarefree simplification follows from Corollary 3.3.

For the degree estimate, $\deg t = 2$ and $\deg q \leq 5$, hence $\deg \Pi \leq 7$. A summand of B indexed by $k \geq 4$ has degree at most

$$2 \deg t + (k - 2) + k \deg q \leq 4 + (k - 2) + 5k = 6k + 2.$$

The preceding terms have degrees at most 1, 6, and 7, so $\deg B \leq 6N + 2$. Similarly, $(xq)^k$ has degree at most $6k$, while the remaining terms of C have degree at most 4. Thus $\deg C \leq 6N$. A linear output normalization does not change these bounds. $\square$

4 Prescribing finite étale algebras

The construction used in the proof can be read from left to right:

$$\boxed{\text{squarefree } P(T) \mapsto G(S) = P(a + S) - P(a) \mapsto \tilde{F}_G^{-1}(y_{P,a}) \simeq \text{Spec } K[T]/(P).}$$

The last arrow is an isomorphism of schemes, not merely a matching of K -points or a count after passing to an algebraic closure.

Theorem 4.1 (Polynomial realization). *Let $P \in K[T]$ be squarefree of degree $N \geq 3$, and let $a \in K$ satisfy*

$$P'(a)P'''(a) \neq 0.$$

Put $G(S) = P(a + S) - P(a)$, let $F_G = (\Pi, B, C)$ be the quadratic-gauge map of (2.14), and set

$$\tilde{F}_G = L \circ F_G, \quad L(\Pi, B, C) = \left(\Pi, -\frac{B}{2}, C \right).$$

Then

$$\det D\tilde{F}_G = 1, \quad \text{gdeg}(\tilde{F}_G) = N.$$

At

$$y_{P,a} = \left(1, 0, -\frac{2P(a)}{P'(a)} \right) \tag{4.1}$$

the full Keller fiber is

$$\boxed{\tilde{F}_G^{-1}(y_{P,a}) \simeq \text{Spec } K[T]/(P).}$$

Proof. The linear and cubic coefficients of G are $P'(a)$ and $P'''(a)/3!$, so the quadratic gauge of Section 2 is defined. At the target (4.1), its inverse polynomial is

$$G(S) - \frac{P'(a)}{2} \left(-\frac{2P(a)}{P'(a)} \right) = P(a + S).$$

Translation preserves squarefreeness, so Corollary 3.3 identifies the fiber with $\text{Spec } K[S]/(P(a + S))$.

Translation back to the original polynomial gives

$$K[S]/(P(a + S)) \longrightarrow K[T]/(P(T)), \quad \bar{S} \longmapsto \bar{T} - a,$$

with inverse $\bar{T} \mapsto \bar{S} + a$. The output map L fixes the chosen target, multiplies the Jacobian by $-1/2$, and changes neither geometric degree nor fiber length. Corollary 3.4 therefore gives $\det D\tilde{F}_G = 1$ and $\text{gdeg}(\tilde{F}_G) = N$. Since the displayed fiber has length N , it is full. $\square$

Corollary 4.2 (Effective and functorial form). *Under the hypotheses of Theorem 4.1,*

$$\max_i \deg(\tilde{F}_G)_i \leq 6N + 2.$$

For every commutative K -algebra R , the scheme isomorphism induces a bijection

$$\mathrm{Hom}_{K\text{-alg}}(K[T]/(P), R) \simeq \{u \in R^3 : \tilde{F}_G(u) = y_{P,a}\},$$

natural in R .

Proof. The degree bound is Corollary 3.4. The natural bijection is the represented-fiber equivalence in Corollary 3.3, composed with the translation $K[S]/(P(a+S)) \simeq K[T]/(P)$. $\square$

Lemma 4.3 (Monogenicity). *Every finite étale algebra over an infinite field is generated by one element. Equivalently, if A has rank N , then*

$$A \simeq K[T]/(P)$$

for a squarefree polynomial $P \in K[T]$ of degree N .

Proof. For $\theta \in A$, let $\chi_\theta(T)$ be the characteristic polynomial of multiplication by θ on the K -vector space A , and set

$$\Delta(\theta) = \mathrm{disc}(\chi_\theta).$$

This is a polynomial function on the affine space underlying A . After extending scalars to a separable closure, identify $A \otimes_K K^{\mathrm{sep}}$ with $(K^{\mathrm{sep}})^N$. In these coordinates,

$$\Delta(\theta_1, \dots, \theta_N) = \prod_{i < j} (\theta_i - \theta_j)^2,$$

so Δ is not the zero polynomial. Since K is infinite, some $\theta \in A$ satisfies $\Delta(\theta) \neq 0$. Its characteristic polynomial has N distinct geometric roots, and its minimal polynomial therefore has degree N . Hence $K[\theta] = A$, with squarefree minimal polynomial P . $\square$

Corollary 4.4 (Finite étale realization). *Every finite étale K -algebra A of rank $N \geq 3$ occurs as a full Keller fiber of a Jacobian-one map*

$$\mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^3$$

of geometric degree N .

More precisely, once an isomorphism $A \simeq K[T]/(P)$ and an element $a \in K$ with $P'(a)P'''(a) \neq 0$ are supplied, the map, target, and fiber isomorphism are given by the explicit formulas of Theorem 4.1. For an abstract A , the monogenicity argument proves that such a presentation exists; it does not choose one canonically.

Proof. Lemma 4.3 supplies a squarefree presentation $A \simeq K[T]/(P)$ of degree N . The polynomials P' and P''' are nonzero in characteristic zero, and K is infinite, so some a avoids the finite zero set of $P'P'''$. Apply Theorem 4.1. $\square$

Lemma 4.5 (Degree two is impossible). *Let K be a field of characteristic zero and let $F : \mathbb{A}_K^m \rightarrow \mathbb{A}_K^m$ be a Keller map. Then*

$$[K(X_1, \dots, X_m) : K(F_1, \dots, F_m)] \neq 2.$$

Proof. A separable quadratic extension is Galois, so this is the classical Galois case of the Jacobian theorem due to Campbell, Razar, and Wright [2, 10, 14]. Appendix B records the scalar-extension and descent argument needed to pass from the usual complex statement to an arbitrary characteristic-zero field. $\square$

Corollary 4.6 (Classification of full Keller fibers). *Let X be a finite K -scheme of positive rank N . Then X is isomorphic to a full Keller fiber of a geometric-degree- N Keller map*

$$F : \mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^3$$

if and only if X is finite étale and $N \neq 2$. Consequently, the possible nonzero ranks of full Keller fibers in $\mathbb{A}_K^3$ are exactly

$$\boxed{1, 3, 4, 5, \dots}.$$

Proof. Every Keller fiber is finite étale. If it is a full Keller fiber of rank N , its map has geometric degree N , so Lemma 4.5 excludes $N = 2$. Conversely, the identity realizes every rank-one finite étale K -scheme, namely $\text{Spec } K$, and Corollary 4.4 realizes every finite étale scheme of rank $N \geq 3$ as a full Keller fiber. Empty and shorter special fibers are Keller fibers, but they are not full Keller fibers and hence do not alter the rank list. $\square$

Remark 4.7 (Categorical reformulation). *For $N \geq 1$, let $\text{FullFib}_{3,N}(K)$ be the replete full subcategory of finite K -schemes whose objects are isomorphic to full Keller fibers of geometric-degree- N maps in $\mathbb{A}_K^3$, and let FinEt_K^N be the full subcategory of rank- N finite étale K -schemes. The corollary is equivalently*

$$\text{FullFib}_{3,N}(K) = \begin{cases} \text{FinEt}_K^N, & N = 1 \text{ or } N \geq 3, \\ \emptyset, & N = 2. \end{cases}$$

This essential image remembers the resulting fiber scheme, not a chosen map–target pair (F, y) ; different pairs may realize the same object. On coordinate algebras the statement is contravariant under Spec .

4.1 Scalar extension and arithmetic transfer

Proposition 4.8 (Compatibility with extension of scalars). *Let $K \hookrightarrow K'$ be an extension of characteristic-zero fields. For a squarefree $P \in K[T]$ and an admissible translation parameter $a \in K$, the map $\tilde{F}_G$ of Theorem 4.1, after coefficientwise extension to K' , is exactly the map obtained from the same P and a over K' . Its distinguished fiber satisfies*

$$\tilde{F}_{G,K'}^{-1}(y_{P,a}) \simeq \text{Spec}((K[T]/(P)) \otimes_K K') \simeq \text{Spec } K'[T]/(P).$$

Proof. Every formula in (2.11)–(2.14), the output normalization L , and the target (4.1) is obtained from the coefficients of P and a by field operations. The construction therefore commutes with scalar extension. The fiber identity is the usual base-change isomorphism for a polynomial quotient, combined with Theorem 4.1. $\square$

Corollary 4.9 (Number fields as connected full Keller fibers). *Every finite separable field extension L/K of degree at least three occurs as a connected full Keller fiber of a Jacobian-one polynomial map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ of coordinate degree at most $6[L : K] + 2$. Given a primitive element and its minimal polynomial, the map is explicit.*

Proof. Apply Theorem 4.1 to the separable minimal polynomial of a chosen primitive element. $\square$

Corollary 4.10 (Arithmetic data are preserved). *Let A be a finite étale algebra of rank at least three over a number field. Any realization supplied by Corollary 4.4 has fiber $\text{Spec } A$ itself. It therefore preserves:*

- *connectedness and field decomposition;*
- *real signature;*
- *splitting field and Galois action on geometric points;*
- *the local algebras $A \otimes_K K_v$ at every place;*
- *for any fixed integral model, factorization types away from its bad primes;*
- *the existence or nonexistence of rational and local points.*

Thus an arithmetic example built as a finite étale algebra of rank at least three can be transferred without alteration to a full Keller fiber. Proposition 4.8 gives the corresponding compatibility with local fields and other scalar extensions.

5 Symmetric monodromy

Lemma 5.1 (Morse slices of the quadratic pencil). *Let k be an algebraically closed field of characteristic zero, and let*

$$H(S) = h_1 S + \cdots + h_N S^N, \quad h_1 h_N \neq 0.$$

Then

$$f_b(S) = H(S) - \frac{h_1 b}{2} S^2$$

is Morse for all but finitely many $b \in k$.

Proof. A repeated root r of

$$H(S) - \frac{h_1}{2}(BS^2 + C)$$

cannot be zero, because $H'(0) = h_1$. Solving $E(r) = E'(r) = 0$ gives

$$B(r) = \frac{H'(r)}{h_1 r}, \quad C(r) = \frac{2H(r) - rH'(r)}{h_1}, \quad r \in \mathbb{G}_m. \quad (5.1)$$

Their differentials satisfy

$$dC = -r^2 dB. \quad (5.2)$$

Let $M = k(B(r), C(r)) \subset k(r)$. Equation (5.2), with B nonconstant, shows that $r^2 \in M$. Since $[k(r) : k(r^2)] = 2$, either $M = k(r^2)$ or $M = k(r)$. The first possibility is excluded by the pole of order one of $B(r)$ at $r = 0$: every element of $k(r^2)$ has even valuation there. Thus $M = k(r)$, so (5.1) is birational onto its image.

The rational map $r \mapsto B(r)$ has degree $N - 1$. Its only poles are the pole of order one at 0 and the pole of order $N - 2$ at infinity; the latter is present because $h_N \neq 0$. Outside finitely many values of b , the fiber $B(r) = b$ therefore consists of $N - 1$ distinct points. They are precisely the critical points of f_b .

At such a point,

$$f_b(r) = H(r) - \frac{rH'(r)}{2} = \frac{h_1}{2} C(r).$$

The normalization map in (5.1) is an isomorphism away from finitely many points of the image curve. After excluding the finitely many B -coordinates of its exceptional points, no two points in a fiber of B have the same C -coordinate. Thus the critical values of f_b are pairwise distinct, and f_b is Morse. $\square$

Lemma 5.2 (Specialization through the ordered-root cover). *Let k be a field, let U be a connected normal k -variety, and let*

$$f(S) \in \Gamma(U, \mathcal{O}_U)[S]$$

*have degree N , with leading coefficient and discriminant invertible on U . Let $G \leq S_N$ be the Galois group of f over $k(U)$, acting on a chosen ordering of its roots. If Z is a connected normal k -variety and $i : Z \rightarrow U$ is a morphism, then the Galois group of i^*f over $k(Z)$, in its root action, is conjugate in S_N to a subgroup of G .*

In particular, let k be algebraically closed, fix $p \in k^$ and $b \in k$, and suppose that $E_{p,b,C}(S)$ is separable over $k(C)$. Then its splitting-field Galois group is conjugate to a subgroup of the Galois group of $E_{\Pi,B,C}(S)$ over $k(\Pi, B, C)$.*

Proof. After dividing by the invertible leading coefficient, form the finite étale U -scheme of ordered N -tuples of pairwise distinct roots. Choose the connected component Y containing the chosen generic ordering. The induced cover $Y \rightarrow U$ is the splitting cover: it is a finite étale G -torsor, and its function field is the splitting field of f .

The pullback $Y_Z = Y \times_U Z$ is again a finite étale G -torsor, although it need not be connected. Choose a connected component Y_0 . The subgroup

$$H = \{g \in G : gY_0 = Y_0\}$$

acts simply transitively on the geometric fibers of $Y_0 \rightarrow Z$. Consequently $k(Y_0)/k(Z)$ is the splitting field of i^*f and $\text{Gal}(k(Y_0)/k(Z)) = H \leq G$. A different ordering or a different component conjugates H in S_N , which proves the first statement.

For the asserted specialization, take

$$U = \text{Spec } k[\Pi, \Pi^{-1}, B, C, \Delta_G^{-1}]$$

and divide $E_{\Pi,B,C}$ by its invertible leading coefficient. Since $\Delta_G(p, b, C)$ is a nonzero polynomial, the open curve

$$Z = \text{Spec } k[C, \Delta_G(p, b, C)^{-1}]$$

is connected and maps to U by $\Pi \mapsto p$, $B \mapsto b$. The first statement gives the required subgroup inclusion. $\square$

Theorem 5.3 (Full symmetric generic monodromy). *For every seed (2.1), not merely for a generic choice of its coefficients, the geometric and arithmetic monodromy groups of F_G are*

$$\boxed{\text{Mon}_{\text{geom}}(F_G) = \text{Mon}_{\text{arith}}(F_G) = S_N.}$$

Equivalently, the splitting field of $E_{\Pi,B,C}(S)$ over $K(\Pi, B, C)$ has Galois group S_N , and the same is true after replacing K by an algebraic closure. The conclusion is unchanged for the Jacobian-one normalization $\tilde{F}_G$.

Proof. We work first over $\overline{K}$. Fix $p \in \overline{K}^*$ and put $H(S) = G_p(S)$. Consider

$$f_b(S) = H(S) - \frac{g_1 b}{2} S^2.$$

Lemma 5.1 shows that f_b is Morse for all but finitely many $b \in \overline{K}$.

Choose one such b . The classical Morse-polynomial theorem [12, Theorem 4.4.5] gives

$$\text{Gal}(f_b(S) - T / \overline{K}(T)) = S_N.$$

Indeed, the polynomial cover is connected, its finite branch inertia groups are transpositions, and those inertia groups generate; a transitive permutation group generated by transpositions is the full symmetric group. After the substitution $T = g_1 C/2$, this is the inverse cover on the line $\Pi = p$, $B = b$. Lemma 5.2 embeds its monodromy into the generic geometric monodromy. Hence

$$S_N \leq \text{Mon}_{\text{geom}}(F_G) \leq S_N,$$

which proves equality. Finally,

$$\text{Mon}_{\text{geom}}(F_G) \triangleleft \text{Mon}_{\text{arith}}(F_G) \leq S_N$$

forces the arithmetic group to be S_N as well. A linear automorphism of the target does not change either monodromy group, so the same conclusion holds for $\widehat{F}_G$. $\square$

Corollary 5.4 (Absolute and stable compositional atomicity). *Let F be any realization map F_G or $\widehat{F}_G$ above. If*

$$F = H_2 \circ H_1$$

as polynomial self-maps of $\mathbb{A}_K^3$, then one factor is a polynomial automorphism. The same conclusion holds after every characteristic-zero extension of K , after every stabilization

$$F \times \text{id}_{\mathbb{A}^r},$$

and after polynomial changes of source and target coordinates. Thus every realization map is absolutely and stably compositionally indecomposable.

Proof. Over an algebraic closure, write

$$K_0 = \overline{K}(F_1, F_2, F_3) \subset L = \overline{K}(x, y, z).$$

The generic sheets form the natural S_N -set. This action is primitive: its point stabilizer S_{N-1} is maximal in S_N . Galois correspondence therefore gives no proper intermediate field between K_0 and L .

Any factorization $F = H_2 \circ H_1$ gives such an intermediate field. Both factors are dominant, and the chain rule

$$\det DF = (\det DH_2) \circ H_1 \cdot \det DH_1 \in K^*$$

shows that both are Keller maps: the only units in a polynomial ring are constants, and dominance of H_1 makes pullback injective. Primitivity forces one factor to have geometric degree one. A degree-one Keller self-map of affine space is an automorphism: it is birational and étale, hence an open immersion by Zariski's Main Theorem; a proper complement is excluded first in codimension one by the unit group of affine space and then in codimension at least two by normal Hartogs extension.

Extension of the constant field does not change the geometric monodromy action. Stabilization replaces $K_0 \subset L$ by

$$K_0(t_1, \dots, t_r) \subset L(t_1, \dots, t_r),$$

with the same Galois closure, group, and point stabilizer. The preceding argument therefore applies after every extension and stabilization. Conjugating by polynomial automorphisms plainly preserves the conclusion. $\square$

Remark 5.5. *Taking $N = 12$ gives an atomic Keller map of composite geometric degree. An explicit sparse formula and independent certificate are recorded separately in the accompanying repository.*

Corollary 5.6 (Hilbertian specializations). *Suppose that K is Hilbertian. For every seed (2.1), there are infinitely many targets in $K^* \times K^2$ whose fibers are connected full finite-étale schemes of rank N and whose splitting fields have Galois group S_N over K .*

Proof. Choose $p \in K^*$. Lemma 5.1 excludes only finitely many values of b in an algebraic closure. A Hilbertian field is infinite, so some $b \in K$ makes f_b Morse. Let L be the splitting field of $f_b(S) - T$ over $K(T)$. Serre's Morse-polynomial theorem gives

$$\text{Gal}(L/K(T)) = S_N$$

[12, Theorem 4.4.5]. Applied after extending the constants to $\overline{K}$, the same theorem gives geometric group S_N . Thus the arithmetic and geometric groups agree, their constant-field quotient is trivial, and $L/K(T)$ is regular over K .

Hilbert irreducibility in its Galois form [12, Chapter 3] now supplies infinitely many $c \in K$ for which $E_{p,b,c}$ remains irreducible and has Galois group S_N . Such a polynomial is squarefree of degree N , so Corollary 3.3 identifies it with a connected full Keller fiber. $\square$

Remark 5.7 (Which hypotheses are used where). *The general localized quadratic-gauge fiber theorem, Theorem 3.2, uses only that π , g_1 , and g_3 are nonzero for the fiber identification, together with the identity*

$$E'(S) = g_1(1 - bS + S\beta(\pi, S) + \pi S^2).$$

Squarefreeness is needed only to conclude that E' is invertible at every root and hence that no localization remains. The condition $g_3 \neq 0$ is what makes the source coefficient g_1/g_3 meaningful. For a translated polynomial $P(a + S) - P(a)$, the cubic coefficient is the third Hasse derivative at a , equal to $P'''(a)/3!$ in characteristic zero.

The transposition inertia in Theorem 5.3 does not contradict the fact that F_G is étale. A repeated inverse root has $E' = 0$, so the reconstruction theorem removes it from the affine source; the corresponding valuation is centered on the nonproperness boundary. Monodromy is taken on the function-field cover, or equivalently on the finite étale restriction over a sufficiently small target open set.

6 Discussion

The core theorem is the polynomial realization theorem. Given a squarefree presentation $K[T]/(P)$ and an admissible a , it gives an explicit Keller map with that full Keller fiber. Monogenicity then yields the corresponding existence statement for an abstract finite étale algebra. The distinction matters: the formulas are explicit from (P, a) , while the primitive-element step supplies no canonical presentation. The classification by rank is a corollary, but it is not the main point: two finite étale

algebras of the same rank can have different field decompositions, Galois actions, signatures, and local behavior, and the construction preserves all of that data.

Several consequences are immediate. A squarefree polynomial of degree at least three becomes an exact inverse polynomial after one translation; an irreducible polynomial gives a connected full Keller fiber; and a degree- N input is realized in $\mathbb{A}^3$ with coordinate degree at most $6N + 2$. Every map produced by the construction has geometric and arithmetic generic monodromy S_N . The primitive action makes every realization map absolutely and stably compositionally indecomposable. Thus monodromy supplies a structural property of the polynomial map, rather than only a generic Galois label. Over a Hilbertian field, each such fixed map has infinitely many connected full Keller fibers with splitting-field group S_N .

The cubic seed admits two useful explanations. Normalized multiplication of a marked linear and quadratic binary form explains its three marked-root branches [13]; the mixed-sign quotient explains its grading and contracted locus [11]. These viewpoints motivate the gauge. The load-bearing argument remains the exact identity $E' = g_1 D$ and the boxed root-to-source reconstruction, which work for every prescribed inverse polynomial and every commutative test algebra.

Appendix A retains the coefficientwise expansion from the compact inverse design to the displayed polynomial coordinates. Appendix B records the scalar-extension and descent argument for the classical degree-two obstruction, and Appendix C gives the verification table linking the main statements to their Lean declarations. Finally, Appendix D contains the logically separate nonproperness calculation: the exact reduced nonproperness locus, the distinguished-target complement, and the complete fiber table on $\Pi = 0$.

This is a transfer statement, not an inverse Galois theorem. It does not say that arbitrary local specifications can always be globalized. It says that once a finite étale algebra with the desired arithmetic properties exists, the Keller construction preserves it exactly. Nor does the monodromy theorem say that one fixed seed realizes every rank- N finite étale algebra: the prescribed algebra is used to choose the seed. Its strengthening is that every resulting cover, including one built for a highly special algebra, has maximal generic monodromy.

Relation to earlier work. To my knowledge, the following prescribed-algebra statement has not appeared before: every finite étale K -algebra of rank at least three occurs as a *full Keller fiber* of a map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$, with explicit formulas once a squarefree presentation and admissible translation are supplied. The rank set $\{1, 3, 4, \dots\}$ follows from this theorem together with the known degree-two obstruction. I searched for earlier results up to 24 July 2026, but this priority statement is necessarily qualified: searches can miss unpublished work or a theorem stated in different language.

The existence of Keller maps in every generic degree is shared terrain. Gallagher gives explicit all-degree families with complete split rational fibers [3]; Migus explicitly uses the available all-degree constructions as input in the real non-dense-image problem [9]. Shaska studies the grading, quotient geometry, and arithmetic thinness of the cubic map [11]. None of these results reverses the quantifiers to realize an arbitrarily prescribed finite étale algebra as the complete fiber. Miranda–Neto studies decompositions of fibers already arising from Keller maps [8], while Lipton and Markakis connect rational images with Hilbert irreducibility [7]. The monodromy argument here uses the classical theorem that a Morse polynomial has symmetric Galois group over the value-function field [12, Theorem 4.4.5].

Provenance and AI assistance. The recent three-dimensional counterexample was announced by Alpöge, who credited Akhil Mathew for suggesting the problem and Fable for producing the example [1]. Its determinant and collision have since received an independent Isabelle/HOL formalization [4];

this paper retains it only as the cubic seed $S + S^3$. Gallagher independently developed an all-degree atlas [3]. Campbell, Razar, and Wright supply the degree-two obstruction [2, 10, 14].

I used generative AI tools during exploration, symbolic-checker development, proof review, literature search, Lean formalization, and manuscript editing. They were useful as fast but fallible collaborators. I checked the stated mathematical claims, rewrote the exposition, and take responsibility for the paper.

A Coefficientwise verification of the polynomial pullback

This appendix verifies that the compact design (2.5)–(2.6) pulls back to the polynomial map (2.14). The calculation treats one generic coefficient g_k at a time; it is not inferred from low-degree examples.

From (2.3) and (2.4),

$$\beta(\Pi, S) = 2\frac{g_2}{g_1}\Pi + \left(3\frac{g_3}{g_1} - 1\right)\Pi S + \sum_{k=4}^N k\frac{g_k}{g_1}\Pi^k S^{k-2}. \quad (\text{A.1})$$

On the source chart,

$$S = \frac{x}{t}, \quad \Pi = tq, \quad Q = y + xq,$$

and therefore, for every $k \geq 4$,

$$\Pi^k S^{k-2} = t^2 x^{k-2} q^k. \quad (\text{A.2})$$

Substitution now gives

$$\begin{aligned} Q + \beta(\Pi, S) &= y + xq + 2\frac{g_2}{g_1}tq + \left(3\frac{g_3}{g_1} - 1\right)xq + \sum_{k=4}^N k\frac{g_k}{g_1}t^2 x^{k-2} q^k \\ &= y + 3\frac{g_3}{g_1}xq + 2\frac{g_2}{g_1}tq + \sum_{k=4}^N k\frac{g_k}{g_1}t^2 x^{k-2} q^k, \end{aligned}$$

which is the displayed coordinate B .

For the third coordinate, write

$$C_* = \frac{2G_\Pi(S)}{g_1} - BS^2.$$

Using $B = Q + \beta(\Pi, S)$ and the definition of β , we obtain

$$C_* = 2S - QS^2 + \left(1 - \frac{g_3}{g_1}\right)\Pi S^3 - \sum_{k=4}^N (k-2)\frac{g_k}{g_1}\Pi^k S^k. \quad (\text{A.3})$$

The terms with $k \geq 4$ already have the desired source form, since

$$\Pi^k S^k = (xq)^k. \quad (\text{A.4})$$

It remains to simplify the cubic part. Put $r = g_3/g_1$. Then

$$\begin{aligned} 2S - QS^2 + (1-r)\Pi S^3 &= \frac{2xt - x^2y - rx^3q}{t^2} \\ &= \frac{x(t+1) - rx^3q}{t^2}, \end{aligned}$$

where the second line uses $xy = t - 1$. Since

$$q = t^2 z + \frac{1}{r} y^2 (1 + 3t),$$

this becomes

$$-rx^3 z + \frac{x(t+1 - (t-1)^2(1+3t))}{t^2}.$$

The elementary identity

$$t+1 - (t-1)^2(1+3t) = t^2(5-3t) \tag{A.5}$$

therefore gives

$$2S - QS^2 + (1-r)\Pi S^3 = x(5-3t) - \frac{g_3}{g_1} x^3 z.$$

Together with (A.3) and (A.4), this is exactly the displayed coordinate C .

We derived the identities after inverting t , but both sides are polynomials in $K[x, y, z]$. Localization at the nonzero polynomial t is injective, so the identities hold everywhere. At fixed Π , their differential has determinant

$$\det \frac{\partial(B, C)}{\partial(S, Q)} = -2 \left(\frac{G'_\Pi(S)}{g_1} - BS \right) = -2D.$$

The source-chart determinant is D^{-1} , and the chain rule gives $\det DF_G = -2$. This is the complete cancellation: each coefficient g_k enters independently, so the same calculation works in every degree.

B The degree-two obstruction over arbitrary characteristic-zero fields

This appendix supplies the scalar-extension and descent details used in Lemma 4.5. Let

$$F = (F_1, \dots, F_m) \in K[X_1, \dots, X_m]^m \quad \text{with} \quad \det \left(\frac{\partial F_i}{\partial X_j} \right) \in K^*,$$

and suppose for contradiction that its function-field degree is two.

The external input. Campbell's unnumbered theorem [2, Theorem, p. 244] states that a polynomial map $H : \mathbb{C}^m \rightarrow \mathbb{C}^m$ has a polynomial inverse if its Jacobian determinant never vanishes and the induced extension

$$\mathbb{C}(H_1, \dots, H_m) \subset \mathbb{C}(X_1, \dots, X_m)$$

is normal. In characteristic zero, normality here is equivalent to the extension being Galois. Razar and Wright later gave algebraic treatments of this Galois case [10, 14]; the argument below needs only Campbell's complex theorem.

Generic degree survives scalar extension. Let $K_0 \subset K$ be the subfield generated over $\mathbb{Q}$ by the finitely many coefficients of the F_i , and write F_0 for the same formulas over K_0 . The nonzero Jacobian determinant makes

$$\varphi_0 : K_0[Y_1, \dots, Y_m] \longrightarrow K_0[X_1, \dots, X_m], \quad Y_i \longmapsto F_i, \tag{B.1}$$

injective and generically finite. Indeed, the Jacobian criterion in characteristic zero makes the F_i algebraically independent, so both fraction fields have transcendence degree m . If their extension

has degree d_0 , there is a nonempty principal open U_0 of the target on which F_0 is finite locally free of rank d_0 : clear denominators in the finite extension and apply generic freeness. For every field extension E/K_0 , base change over $(U_0)_E$ is again finite locally free of rank d_0 . Taking generic fibers gives

$$[E(X_1, \dots, X_m) : E(F_1, \dots, F_m)] = d_0. \quad (\text{B.2})$$

Taking $E = K$ shows that $d_0 = 2$.

The quadratic extension is Galois. Because K_0 is finitely generated over $\mathbb{Q}$, choose an embedding $K_0 \hookrightarrow \mathbb{C}$. Equation (B.2) with $E = \mathbb{C}$ says that

$$\mathbb{C}(X_1, \dots, X_m) / \mathbb{C}(F_1, \dots, F_m)$$

has degree two. It is separable, and every separable quadratic field extension is normal: the second root of the minimal polynomial of an element is its trace minus that element. The complexified map therefore satisfies Campbell's hypotheses and is a polynomial automorphism.

Descent of the inverse. The base change of (B.1) along $K_0 \hookrightarrow \mathbb{C}$ is an isomorphism. Since $\mathbb{C}$ is faithfully flat over K_0 , tensoring the kernel and cokernel of φ_0 with $\mathbb{C}$ detects their vanishing. Hence φ_0 is an isomorphism. After base change to K , the original F is a polynomial automorphism, so its function-field degree is one, contrary to the assumption.

C Verification and Lean correspondence

The table records the public endpoint declarations corresponding to the formalized statements used in the paper. Several rows list more than one declaration because the paper packages construction, naturality, rank, and geometric degree into a single result. Auxiliary implementation lemmas are omitted.

Paper statement	Lean declaration
General localized quadratic-gauge fiber theorem, Theorem 3.2, and its squarefree corollary, Corollary 3.3	<code>generalGaugeLocalizedJacobianOneRepresentingEquiv</code> <code>generalGaugeLocalizedJacobianOneRepresentingEquiv_natural</code>
Jacobian-one effective form, Corollary 3.4	<code>jacobianDet_generalGaugeJacobianOneMap</code> <code>generalGaugeJacobianOneMap_totalDegree</code>
Polynomial realization, Theorem 4.1, and its effective and functorial form, Corollary 4.2	<code>automaticRealization_pageOne</code> <code>automaticJacobianOneFiberRepresentingEquiv_natural</code>
Abstract finite-étale realization, Corollary 4.4	<code>finiteEtalePresentation</code> <code>abstractFiniteEtaleFiberRepresentingEquiv_natural</code> <code>abstractFiniteEtale_pageOne</code>
Compatibility with extension of scalars, Proposition 4.8	<code>realizationMapTarget_map</code> <code>adjoinRootBaseChangeEquiv</code>

D The exact nonproperness locus

The prescribed-realization, classification, monodromy, and atomicity results in the main text do not require the calculation below. We record it separately because it gives an intrinsic geometric description of fullness for the quadratic gauge.

Work geometrically over $\overline{K}$. For a polynomial map $F : \mathbb{A}_{\overline{K}}^n \rightarrow \mathbb{A}_{\overline{K}}^n$, let S_F be the set of target points at which F is not finite over any Zariski neighborhood. Since source and target are affine, this is the same as failure of properness over every neighborhood. Equivalently, compactify the graph in $\mathbb{P}_{\overline{K}}^n \times \mathbb{A}_{\overline{K}}^n$; then S_F is the image of its boundary. We give S_F its reduced closed-subscheme structure. Over $\mathbb{C}$ this agrees with the usual analytic Jelonek set, defined by sequences escaping to infinity.

For a polynomial h , write $V_{\text{red}}(h)$ for the reduced closed subscheme $V(\sqrt{(h)})$. Put

$$\Delta_G(\Pi, B, C) = \text{Disc}_S(E_{\Pi, B, C}(S)). \quad (\text{D.1})$$

Here the discriminant is the universal degree- N polynomial expression in the coefficients of $E_{\Pi, B, C}$. We do not replace it on a special slice by the discriminant of the lower-degree specialization.

Lemma D.1 (Base change of the graph-boundary locus). *Let $k \subset k'$ be a field extension and let $F : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ be a polynomial map. Let $\overline{\Gamma}_F$ be the scheme-theoretic closure of its graph in $\mathbb{P}_k^n \times \mathbb{A}_k^m$, put*

$$Z_F = \overline{\Gamma}_F \cap ((\mathbb{P}_k^n \setminus \mathbb{A}_k^n) \times \mathbb{A}_k^m),$$

and give the image of the proper morphism $Z_F \rightarrow \mathbb{A}_k^m$ its reduced closed-subscheme structure S_F . Then

$$Z_{F_{k'}} = Z_F \times_k k', \quad |S_{F_{k'}}| = |(S_F)_{k'}|.$$

In particular, whether a geometric target point lies in the graph-boundary locus is invariant under extension of algebraically closed fields.

Proof. Field extension is faithfully flat. On each standard affine chart of $\mathbb{P}_k^n \times \mathbb{A}_k^m$, the ideal of the graph closure is the saturation $I : x_0^\infty$ of the graph ideal by the equation x_0 of the hyperplane at infinity. Since the coordinate rings are noetherian, this saturation equals $I : x_0^r$ for some r ; flatness gives

$$(I : x_0^r) \otimes_k k' = (I \otimes_k k') : x_0^r.$$

Thus scheme-theoretic graph closure commutes with the field extension. Intersecting with $x_0 = 0$ also commutes with base change, which proves the identity for Z_F .

For the image statement, let y' be a point of $\mathbb{A}_{k'}^m$, with image y in $\mathbb{A}_k^m$. The fiber $(Z_F)_{y'}$ is obtained from $(Z_F)_y$ by a field extension. A finite-type scheme over a field is empty if and only if it becomes empty after a field extension. Hence $(Z_F)_{y'}$ is nonempty exactly when $(Z_F)_y$ is nonempty. Since a proper morphism has closed image, these fiberwise equivalences give the asserted equality of underlying closed sets. $\square$

Lemma D.2 (Fiber cardinality and nonproperness in characteristic zero). *Let k be an algebraically closed field of characteristic zero and let $F : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ be a dominant étale polynomial map of geometric degree d . Then*

$$y \notin S_F \iff \#F^{-1}(y) = d. \quad (\text{D.2})$$

Proof. Jelonek proves this multiplicity criterion for complex polynomial maps [5, Proposition 6]. We record why that stated complex result is sufficient here. Since F is étale, its Jacobian determinant is a nonzero unit of $k[x_1, \dots, x_n]$, hence a nonzero constant. Descend the coefficients of F , this constant Jacobian identity, and the coordinates of y to a finitely generated field $k_0/\mathbb{Q}$. The resulting map F_0 is étale and therefore dominant; its geometric degree is still d . We also obtain $y_0 \in \mathbb{A}^n(k_0)$ whose base changes are y .

Choose an embedding $k_0 \hookrightarrow \mathbb{C}$ and an algebraic closure Ω of k_0 . The two embeddings of k_0 , into k and into $\mathbb{C}$, extend to embeddings

$$\Omega \hookrightarrow k, \quad \Omega \hookrightarrow \mathbb{C}.$$

Lemma D.1 shows that membership of y_0 in the graph-boundary locus has the same truth value after each of these scalar extensions. The fiber of an étale morphism is finite and reduced; its number of geometric points is unchanged by an extension of algebraically closed fields. The degree of the generic finite separable extension is likewise invariant under scalar extension. We may therefore test both sides of (D.2) after base change to $\mathbb{C}$. There Proposition 6 of [5], with every local multiplicity equal to one, gives the equivalence. $\square$

Lemma D.3 (Exact Π -adic discriminant order). *Assume $N \geq 4$, put $r = N - 3$, and let*

$$R = \overline{K}[B, C], \quad L = \text{Frac}(R).$$

For the degree- N polynomial $E = E_{\Pi, B, C}(S)$, viewed in $L((\Pi))[S]$, extend $\text{ord}_{\Pi}(\Pi) = 1$ to an algebraic closure of $L((\Pi))$. The roots split into three Newton blocks:

number of roots	$\text{ord}_{\Pi}(s)$	residual polynomial
2	0	$BZ^2 - 2Z + C$
1	-1	$Z - \frac{g_1 B}{2g_3}$
r	$-\frac{r+2}{r}$	$g_3 + g_N Z^r$.

(D.3)

All three residual polynomials are separable over L . Consequently

$$\text{ord}_{\Pi} \text{Disc}_S(E) = N^2 - 3N - 2,$$

and, in $R[\Pi]$,

$$\text{Disc}_S(E) = \Pi^{N^2 - 3N - 2} D_G, \quad D_G(0, B, C) = \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1} B^2 (1 - BC).$$

Proof. The coefficient valuations give the lower Newton polygon

$$(0, 0) \longrightarrow (2, 0) \longrightarrow (3, 1) \longrightarrow (N, N).$$

Its slopes are 0, 1, and $(r+2)/r$, with horizontal lengths 2, 1, and r . Rescaling S along each edge gives exactly the three residual polynomials in (D.3). The first has discriminant $4(1 - BC) \neq 0$ in L , the second is linear, and the third is separable because $\text{char } L = 0$ and $g_3 g_N \neq 0$.

Write the roots in the three blocks as A_1, A_2, T , and $V_1, \dots, V_r$. Roots in different blocks have different valuations, so the valuation of their difference is the smaller of the two valuations. Within the first and third blocks, distinct roots of the residual polynomial have distinct initial terms. Thus

$$\text{ord}_{\Pi}(A_1 - A_2) = 0, \quad \text{ord}_{\Pi}(V_i - V_j) = -\frac{r+2}{r} \quad (i \neq j).$$

This is the precise use of residual separability. It proves equality in each entry of the following ledger:

pair type	number of pairs	$\text{ord}_{\Pi}(s_i - s_j)$	contribution to $\text{ord}_{\Pi} \prod_{i < j} (s_i - s_j)^2$
$A - A$	1	0	0
$A - T$	2	-1	-4
$A - V$	$2r$	$-(r+2)/r$	$-4(r+2)$
$T - V$	r	$-(r+2)/r$	$-2(r+2)$
$V - V$	$r(r-1)/2$	$-(r+2)/r$	$-(r-1)(r+2)$.

The leading coefficient of E is $g_N \Pi^N$, so its contribution to

$$\text{Disc}(E) = \text{lc}(E)^{2N-2} \prod_{i < j} (s_i - s_j)^2$$

is $2N(N-1)$. Adding the last column gives

$$2N(N-1) - 4 - 4(r+2) - 2(r+2) - (r-1)(r+2) = N^2 - 3N - 2.$$

It remains only to identify the first nonzero coefficient. Let α_1, α_2 be the roots of $BZ^2 - 2Z + C$, put $b_0 = g_1 B / (2g_3)$, and let $v_1, \dots, v_r$ be the roots of $g_3 + g_N Z^r$. Taking the initial term in every factor of the root-product formula gives

$$\begin{aligned} & \pm g_N^{2N-2} (\alpha_1 - \alpha_2)^2 b_0^4 \left(\prod_j v_j \right)^6 \prod_{i < j} (v_i - v_j)^2 \\ &= \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1} B^2 (1 - BC), \end{aligned}$$

because

$$(\alpha_1 - \alpha_2)^2 = \frac{4(1 - BC)}{B^2}, \quad \left(\prod_j v_j \right)^6 = \left(\frac{g_3}{g_N} \right)^6, \quad \prod_{i < j} (v_i - v_j)^2 = \pm r^r \left(\frac{g_3}{g_N} \right)^{r-1}.$$

This coefficient is nonzero in R . Since Π -adic order in the domain $R[\Pi]$ is unchanged after passage to $L[\Pi]$, the asserted factorization and specialized coefficient follow. $\square$

Theorem D.4 (Exact reduced quadratic-gauge nonproperness). *For every seed (2.1),*

$$\boxed{S_{FG} = V_{\text{red}}(\Delta_G)}. \quad (\text{D.4})$$

Equivalently, their sets of geometric points satisfy $|S_{FG}| = V_{\overline{K}}(\Delta_G)$. More precisely, the following statements hold.

1. *If $\pi \neq 0$ and the multiplicity partition of $E_{\pi, b, c}$ is $(m_1, \dots, m_s)$, then*

$$\#F_G^{-1}(\pi, b, c) = \#\{i : m_i = 1\}, \quad N - \#F_G^{-1}(\pi, b, c) = \sum_{m_i \geq 2} m_i. \quad (\text{D.5})$$

2. *On $\Pi = 0$, the complete geometric fiber table is*

<i>target stratum</i>	$\#F_G^{-1}(0, B, C)$	<i>lost sheets</i>
$B \neq 0, BC \neq 1$	3	$N - 3$
$B \neq 0, BC = 1$	1	$N - 1$
$B = 0$	1	$N - 1$.

(D.6)

3. *For $N = 3$, the raw resultant has an extraneous leading-coefficient factor:*

$$\text{Res}_S(E, E') = \text{unit} \cdot \Pi \Delta_G, \quad \Delta_G(0, B, C) = \frac{g_1^4}{4} B^2 (1 - BC).$$

Hence the entire plane $\Pi = 0$ is not a nonproperness component in degree three.

4. For $N \geq 4$, put

$$e_N = N^2 - 3N - 2.$$

Then

$$\Delta_G = \Pi^{e_N} D_G, \quad D_G(0, B, C) = \kappa_G B^2(1 - BC), \quad \kappa_G \in K^*. \quad (\text{D.7})$$

In particular,

$$S_{F_G} = V_{\text{red}}(\Pi D_G),$$

whose underlying set is $V(\Pi) \cup V(D_G)$, and the resultant and discriminant define the same reduced closed subscheme.

Proof. Because F_G is étale, every geometric fiber is finite and reduced. Its generic cardinality is N , so Lemma D.2 applies with $d = N$.

Suppose first that $\pi \neq 0$. Theorem 3.2 identifies the fiber with

$$\text{Spec}((\overline{K}[S]/(E_{\pi,b,c}))[1/\overline{E'_{\pi,b,c}}]).$$

Thus a distinct root contributes one affine point exactly when it is simple. This proves (D.5); it also shows that the fiber has cardinality N exactly when $\Delta_G(\pi, b, c) \neq 0$.

It remains to analyze $\Pi = 0$, where the inverse chart used above is not available. In the source, $\Pi = tq$. On $q = 0$, one has

$$B = y, \quad C = 2S - BS^2, \quad S = \frac{x}{t}.$$

The residual inverse equation is therefore

$$BS^2 - 2S + C = 0. \quad (\text{D.8})$$

A root reconstructs on $q = 0$ precisely when $1 - BS \neq 0$. Hence this divisor contributes two points when $B \neq 0$ and $BC \neq 1$, no point when $B \neq 0$ and $BC = 1$, and one point when $B = 0$.

On $t = 0$, the relation $1 + xy = 0$ gives $y = -1/x$, while

$$q = \frac{g_1}{g_3} y^2, \quad B = -2y = \frac{2}{x}.$$

Thus $t = 0$ contributes one point when $B \neq 0$ and none when $B = 0$. Once x and y are fixed, the equation for C is linear in z with nonzero coefficient, so the point is unique. The divisors $t = 0$ and $q = 0$ do not meet. This proves (D.6).

For $N = 3$, specializing the degree-three discriminant at $\Pi = 0$ gives

$$\Delta_G(0, B, C) = \frac{g_1^4}{4} B^2(1 - BC).$$

Its zero set is exactly the two short-fiber strata in (D.6). Since the leading coefficient is $g_3 \Pi$, the resultant has one additional factor Π .

Now let $N \geq 4$ and put $r = N - 3$. Lemma D.3 gives the global polynomial identity

$$\Delta_G = \Pi^{e_N} D_G, \quad D_G(0, B, C) = \kappa_G B^2(1 - BC), \quad \kappa_G = \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1} \neq 0.$$

In particular, e_N is the exact global Π -adic order. After specializing (B, C) , the pointwise order is e_N when $B(1 - BC) \neq 0$ and may jump on the two displayed special strata.

No general assertion about discriminants under a drop of leading coefficient is being used. For $N = 3$, the explicit specialized formula cuts out exactly the two short-fiber strata. For $N \geq 4$, Δ_G vanishes on the whole slice $\Pi = 0$, and every fiber on that slice has fewer than N points by (D.6). On $\Pi \neq 0$, the ordinary degree- N discriminant detects exactly the repeated-root, hence short-fiber, locus. The fiber table and (D.2) now show that the zero set of Δ_G is exactly the underlying nonproperness set, and taking radicals gives (D.4). $\square$

Corollary D.5 (Intrinsic fullness of the distinguished fiber). *For the seed $G(S) = P(a + S) - P(a)$ used in Theorem 4.1, the distinguished target $y_{P,a}$ lies in the maximal open subset on which $\tilde{F}_G$ is finite étale of degree N .*

Proof. At $y_{P,a}$, the inverse polynomial is $P(a + S)$. Hence

$$\Delta_G(y_{P,a}) = \text{Disc}(P(a + S)) = \text{Disc}(P) \neq 0.$$

Theorem D.4 places the target outside the nonproperness hypersurface. Proper and étale is finite étale, and the degree is N . $\square$

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